

Multi-Source Candidate Data Transformer — Technical Design

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Pipeline

detect → **extract** → **normalize** → **merge** → **confidence** → **project** → **validate**. Detect dispatches each manifest entry to the right extractor. Extract pulls a loosely-canonical raw record out of each source (this is where ATS's own field names get remapped onto our concepts), wrapped so a bad source produces a logged error, never a crash. Normalize converts per-field raw values into our canonical formats. Merge combines normalized values across sources into one record with conflict resolution. Confidence scores every populated field. Project reshapes the canonical record per the runtime config. Validate checks the projected output's shape before returning it.

Canonical Schema & Normalized Formats

Default schema matches the brief's table (candidate_id, full_name, emails[], phones[], location, links, headline, years_experience, skills[], experience[], education[], provenance[], overall_confidence), plus **source_errors** (which sources were skipped and why — answers "why is this profile thin?" the same way provenance answers "why is this value what it is?").

- **Phones** → E.164 (*phonenumbers*). No country code + no location hint → falls back to a configured default region, but flagged as a guess (lower confidence), never presented as a clean parse.
- **Dates** → YYYY-MM. Year-only ("2019") → 2019-01 with an explicit low-precision note, not a confident-looking fake month. "Present"/"Current" → open-ended (null end), not coerced into a real date.
- **Country** → ISO 3166-1 alpha-2 (*pycountry* + alias table). 2-letter tokens are checked against US state codes *before* country lookup, so "San Francisco, CA" doesn't resolve to Canada.
- **Skills** → canonical name via an alias table ("js"/"JS" → JavaScript). Unknown skills are kept as-is with reduced confidence rather than dropped.
- **Names** → trimmed/whitespace-collapsed only, never re-cased — auto-titlecasing breaks real names ("O'Brien") and is exactly the confident-but-wrong move the brief warns against.

Merge & Conflict-Resolution Policy

Match key: an upstream manifest supplies the crosswalk (which CSV row / ATS record / GitHub fixture / notes file belongs to a given candidate_id) — see Descoped.

Scalar fields (name, headline, years_experience, location): winner-take-all by a per-field source-priority table (e.g. ATS > CSV > GitHub > notes for title; GitHub > notes > ATS for skills, since code is more objective evidence than a recruiter's guess). $\text{field_confidence} = \text{source_weight} \times \text{normalization_quality}$; agreeing sources add a bonus (capped at 1.0); disagreeing ones cost a penalty. One strong source beats two weak ones that agree with each other — majority vote among low-priority sources does not override a single high-priority one. Every losing value is kept in provenance, never discarded.

List fields (emails, phones, skills, experience, education): unioned and deduplicated, not winner-take-all — a candidate can legitimately have two emails, not two "true" current titles. **overall_confidence** = sum of populated-field confidences ÷ a *fixed* field count, so a thin profile scores low rather than being inflated by the few confident fields it happens to have.

Runtime Config (Projection Layer)

The full canonical record is always built first; the config is a pure downstream lens with no special casing — the **default schema runs through the same projection code path** as any custom config, which is what makes "no code changes" structural rather than a promise. Each field spec has *path* (output key), *from* (canonical path, supports emails[0] / skills[].name addressing), *type*, optional *required* and *normalize* (override the display form of an already-normalized value, e.g. E.164 → national). Top-level toggles: *include_confidence* (also strips per-skill confidence), *include_provenance* (also gates source_errors), *on_missing* ∈ {null, omit, error}. **error** fails loud for that one candidate in a batch — it does not take down the rest.

Edge Cases Handled

- **Missing/corrupt source** (malformed JSON, missing file, empty notes) — logged in source_errors and skipped; candidate still gets a profile from whatever survives.
- **Conflicting values** — priority-based winner; loser kept in provenance with the reason.
- **Partial/messy dates** — best-effort to YYYY-MM; year-only flagged low-precision, never a fabricated month.
- **Phone with no country code or location hint** — default-region guess, explicitly flagged, or null if it still doesn't validate — never invented.
- **Same skill, different casing/spelling** — canonicalized and merged into one entry with combined sources.

Deliberately Out of Scope (time-boxed)

Fuzzy cross-source identity resolution with no shared ID (its own ML/heuristics project — pushed to the manifest layer here); resume PDF/DOCX parsing (used GitHub + notes as the unstructured pair instead; the extractor interface is pluggable so a resume extractor is one more module, not a redesign); live GitHub API calls in the test suite (cached fixture used instead, for deterministic/offline runs — the live-fetch path exists but isn't exercised by tests); NER-based skill extraction from free text (dictionary + keyword matching only); any UI beyond a CLI.